DATABASE DESIGN & REFERENCE

SmartFleet

Schema V1 | PostgreSQL | October 2026

System architecture, entity relationships, operational rules and developer guide

Source of truth: SQL migrations 001–022 supplied with this project. Validation evidence: manual integrity checks and corrected automated suite (18 PASS, 0 FAIL), run by the project owner.

At a glance

28 tables	**40 documented FK links***	**22 schema migrations**

* Includes the transfer_id relationship introduced in migration 022; relationship count is extracted from SQL source and may omit programmatic/deferred link checks.

1. The database in plain English

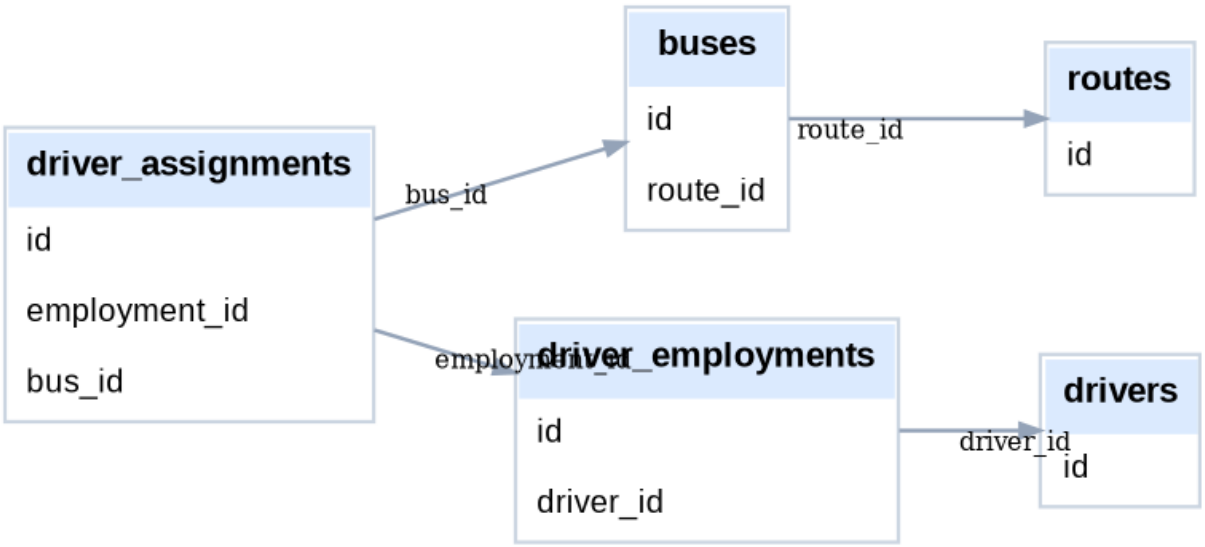
SmartFleet is a digital record book for a small bus transport business. It stores which buses and drivers the business has, who was assigned to a bus and when, how much cash the owner received, what the business spent, and what it owes or is owed. The money records connect back to specific events: for example, a bus cashing collection must agree with the cash received in the account; a driver salary payment must agree with the ledger and any advances recovered; and a bank transfer must account for both sides and its fee. PostgreSQL holds these records in linked tables and checks the important rules automatically, reducing the risk of missing, duplicated or contradictory financial entries. The first application is a Streamlit web interface, but the database is designed to remain usable if the interface later changes to FastAPI and React.

How to read the diagrams

Each rectangle is a database table. Arrows point from the table holding the foreign key to the referenced table. Labels on arrows identify the referencing column. Diagrams are grouped to stay legible on a standard printed page; certain shared tables are repeated for context. Some relationships are enforced by integrity triggers rather than ordinary foreign keys, and are described in the rules section.

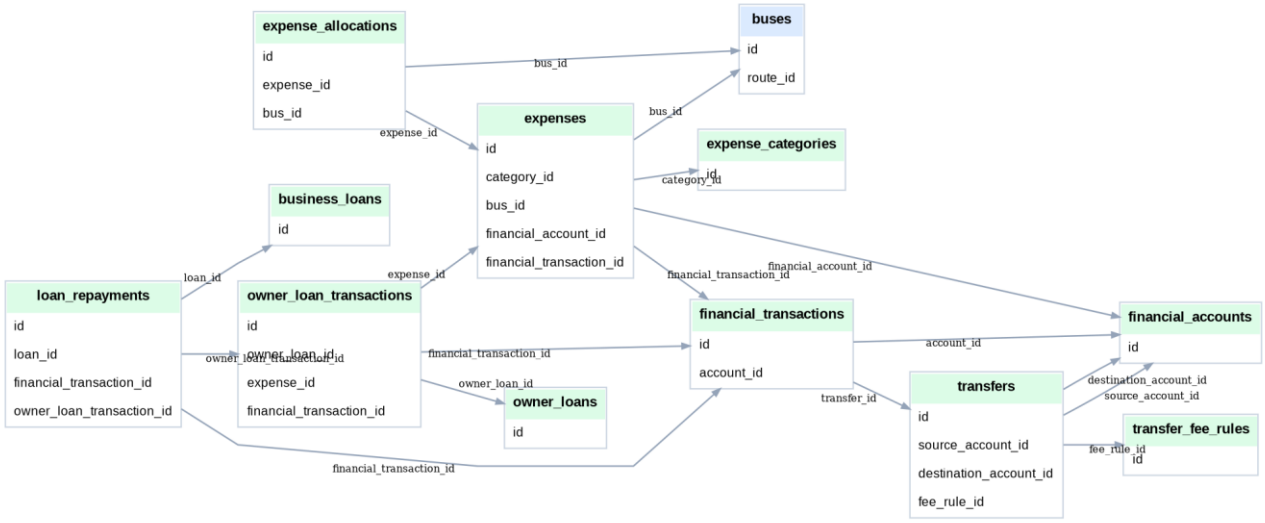
2. Architecture and relationship diagrams

Fleet & people



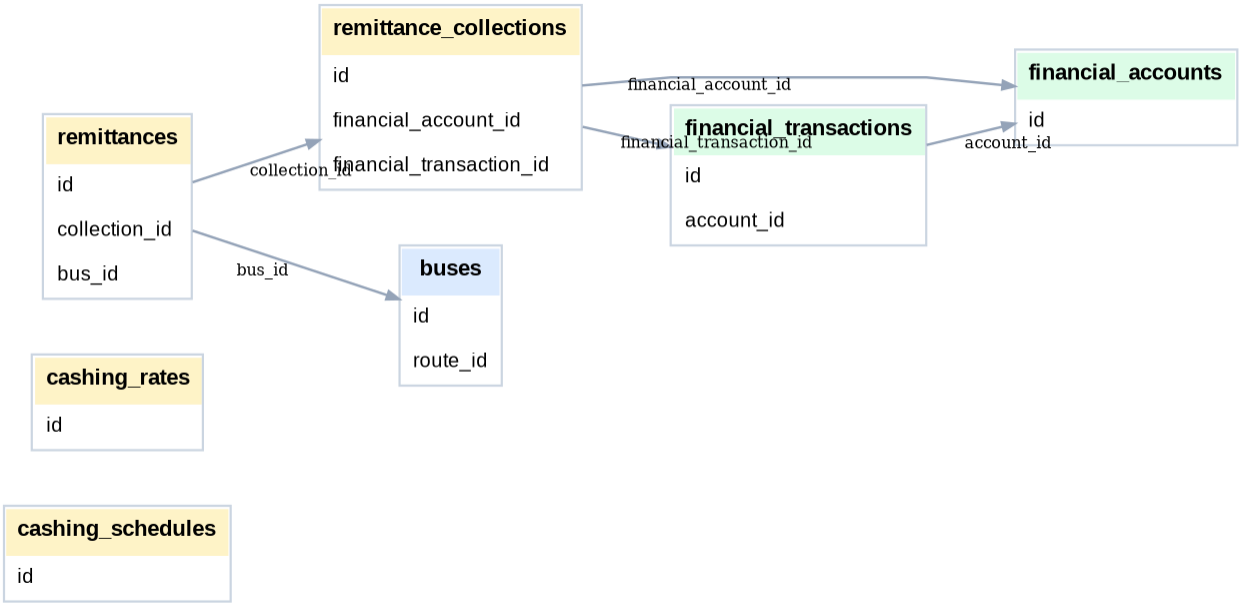
Relationship diagram — Fleet & people

Money & liabilities



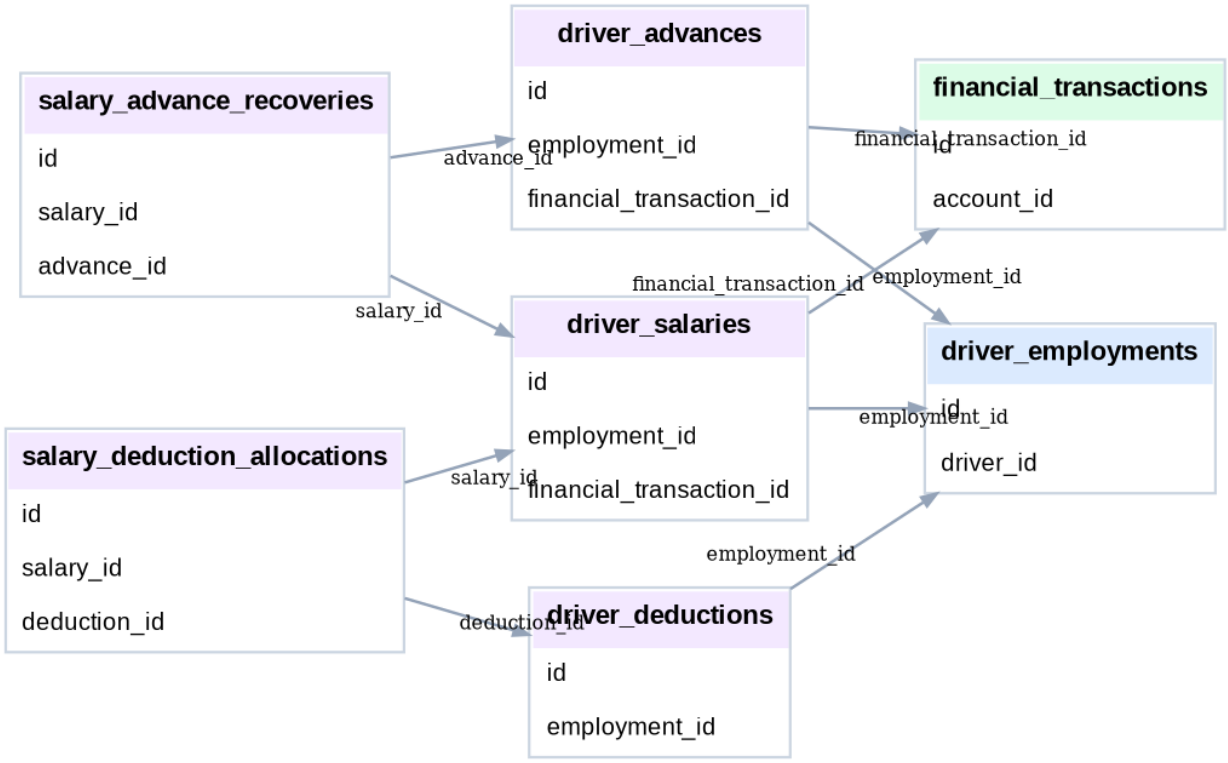
Relationship diagram — Money & liabilities

Revenue & remittances



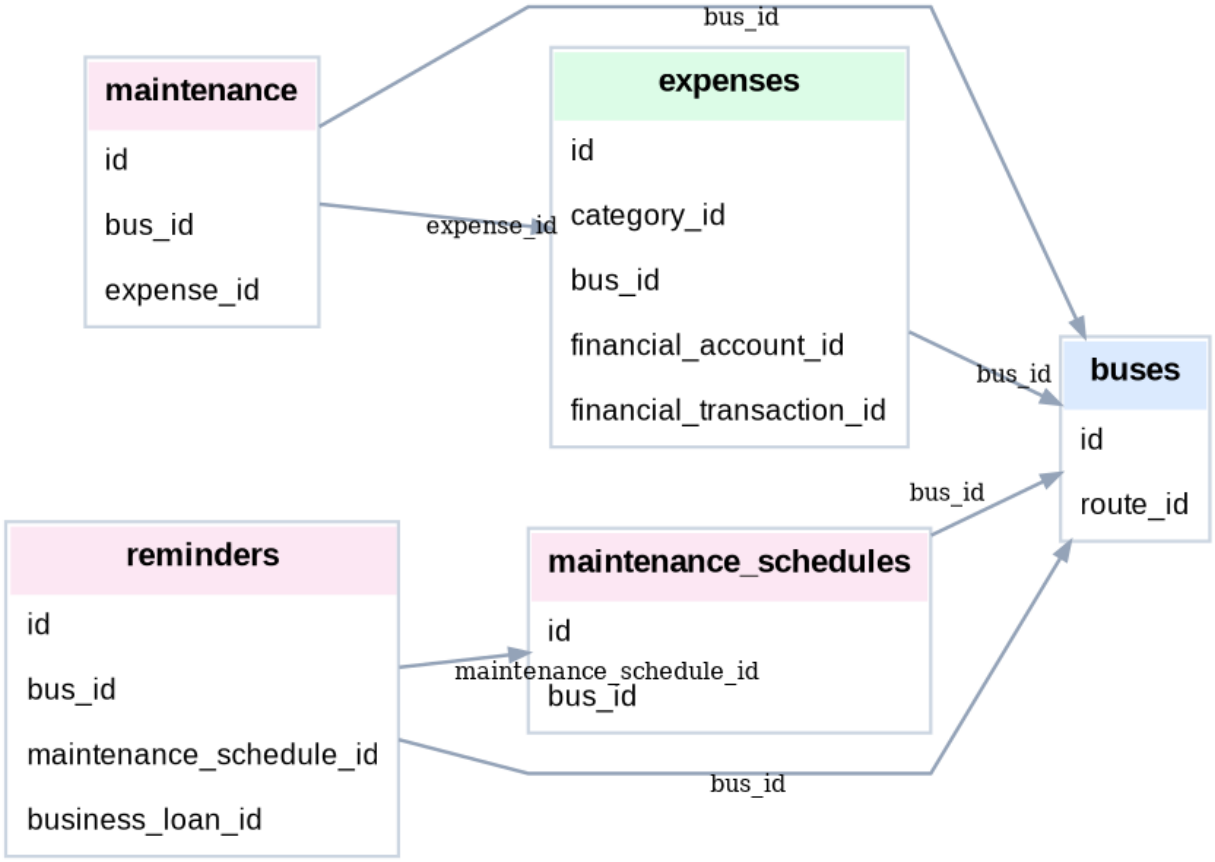
Relationship diagram — Revenue & remittances

Drivers & payroll



Relationship diagram — Drivers & payroll

Maintenance & reminders



Relationship diagram — Maintenance & reminders

3. Table catalogue

The definitions below list the original creation migration and column definitions, including defaults, generated values and nullable fields where available. Migration 022 adds additional enforcement, which is summarised in the integrity section. PK means primary key; FK means foreign key.

Fleet & people

routes

Introduced in 001 — 001_create_routes.sql

Column	Definition (from CREATE TABLE)
id	INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY
description	TEXT
active	BOOLEAN NOT NULL DEFAULT TRUE
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

Relationships: No outgoing foreign keys in source definition.

buses

Introduced in 002 — 002_create_buses.sql

Column	Definition (from CREATE TABLE)
id	INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY
route_id	INTEGER NOT NULL
date_acquired	DATE
notes	TEXT
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

Relationships: route_id → routes.id

drivers

Introduced in 003 — 003_create_drivers.sql

Column	Definition (from CREATE TABLE)
id	INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY
address	TEXT
notes	TEXT
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

Relationships: No outgoing foreign keys in source definition.

driver_employments

Introduced in 004 — 004_create_driver_employments.sql

Column	Definition (from CREATE TABLE)
id	INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY
driver_id	INTEGER NOT NULL
start_date	DATE NOT NULL
end_date	DATE
notes	TEXT
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

Relationships: driver_id → drivers.id

driver_assignments

Introduced in 005 — 005_create_driver_assignments.sql

Column	Definition (from CREATE TABLE)
id	INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY
employment_id	INTEGER NOT NULL
bus_id	INTEGER NOT NULL
start_date	DATE NOT NULL
end_date	DATE
notes	TEXT
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

Relationships: employment_id → driver_employments.id; bus_id → buses.id

Money & liabilities

financial_accounts

Introduced in 006 — 006_create_financial_accounts.sql

Column	Definition (from CREATE TABLE)
id	INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY
active	BOOLEAN NOT NULL DEFAULT TRUE
notes	TEXT
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

Relationships: No outgoing foreign keys in source definition.

financial_transactions

Introduced in 007 — 007_create_financial_transactions.sql

Column	Definition (from CREATE TABLE)
id	INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY
account_id	INTEGER NOT NULL
transaction_date	DATE NOT NULL
description	TEXT
notes	TEXT
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

Relationships: account_id → financial_accounts.id; transfer_id → transfers.id

transfers

Introduced in 008 — 008_create_transfers.sql

Column	Definition (from CREATE TABLE)
id	INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY
transfer_date	DATE NOT NULL
source_account_id	INTEGER NOT NULL
destination_account_id	INTEGER
fee_rule_id	INTEGER
notes	TEXT
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

Relationships: source_account_id → financial_accounts.id; destination_account_id → financial_accounts.id; fee_rule_id → transfer_fee_rules.id

transfer_fee_rules

Introduced in 008 — 008_create_transfers.sql

Column	Definition (from CREATE TABLE)

id	INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY
active	BOOLEAN NOT NULL DEFAULT TRUE
notes	TEXT
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

Relationships: No outgoing foreign keys in source definition.

expense_categories

Introduced in 010 — 010_create_expenses.sql

Column	Definition (from CREATE TABLE)
id	INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY
active	BOOLEAN NOT NULL DEFAULT TRUE
notes	TEXT
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

Relationships: No outgoing foreign keys in source definition.

expenses

Introduced in 010 — 010_create_expenses.sql

Column	Definition (from CREATE TABLE)
id	INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY
expense_date	DATE NOT NULL
category_id	INTEGER NOT NULL
description	TEXT NOT NULL
bus_id	INTEGER
financial_account_id	INTEGER
financial_transaction_id	INTEGER
notes	TEXT
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

Relationships: category_id → expense_categories.id; bus_id → buses.id; financial_account_id → financial_accounts.id; financial_transaction_id → financial_transactions.id

expense_allocations

Introduced in 011 — 011_create_expense_allocations.sql

Column	Definition (from CREATE TABLE)
id	INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY
expense_id	INTEGER NOT NULL
bus_id	INTEGER NOT NULL
notes	TEXT
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

Relationships: expense_id → expenses.id; bus_id → buses.id

owner_loans

Introduced in 012 — 012_create_owner_loans.sql

Column	Definition (from CREATE TABLE)
id	INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY
active	BOOLEAN NOT NULL DEFAULT TRUE
notes	TEXT
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

Relationships: No outgoing foreign keys in source definition.

owner_loan_transactions

Introduced in 012 — 012_create_owner_loans.sql

Column	Definition (from CREATE TABLE)
id	INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY
owner_loan_id	INTEGER NOT NULL
transaction_date	DATE NOT NULL
description	TEXT NOT NULL
expense_id	INTEGER
financial_transaction_id	INTEGER
notes	TEXT
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

Relationships: owner_loan_id → owner_loans.id; expense_id → expenses.id; financial_transaction_id → financial_transactions.id

business_loans

Introduced in 013 — 013_create_business_loans.sql

Column	Definition (from CREATE TABLE)
id	INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY
start_date	DATE NOT NULL
term_months	INTEGER
notes	TEXT
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

Relationships: No outgoing foreign keys in source definition.

loan_repayments

Introduced in 013 — 013_create_business_loans.sql

Column	Definition (from CREATE TABLE)
id	INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY
loan_id	INTEGER NOT NULL
repayment_date	DATE NOT NULL
financial_transaction_id	INTEGER
owner_loan_transaction_id	INTEGER
notes	TEXT
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

Relationships: loan_id → business_loans.id; financial_transaction_id → financial_transactions.id; owner_loan_transaction_id → owner_loan_transactions.id

Revenue & remittances

cashing_schedules

Introduced in 014 — 014_create_remittances.sql

Column	Definition (from CREATE TABLE)
id	INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY
day_of_week	INTEGER NOT NULL
active	BOOLEAN NOT NULL DEFAULT TRUE
notes	TEXT
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

Relationships: No outgoing foreign keys in source definition.

cashing_rates

Introduced in 021 — 021_add_cashing_rate_configuration.sql

Column	Definition (from CREATE TABLE)
id	INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY
active	BOOLEAN NOT NULL DEFAULT TRUE
effective_from	DATE NOT NULL DEFAULT CURRENT_DATE
effective_to	DATE
notes	TEXT
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

Relationships: No outgoing foreign keys in source definition.

remittance_collections

Introduced in 014 — 014_create_remittances.sql

Column	Definition (from CREATE TABLE)
id	INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY
collection_date	DATE NOT NULL
financial_account_id	INTEGER NOT NULL
financial_transaction_id	INTEGER
notes	TEXT
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

Relationships: financial_account_id → financial_accounts.id; financial_transaction_id → financial_transactions.id

remittances

Introduced in 014 — 014_create_remittances.sql

Column	Definition (from CREATE TABLE)
id	INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY
collection_id	INTEGER NOT NULL
bus_id	INTEGER NOT NULL
normal_days	INTEGER NOT NULL DEFAULT 0

sunday_days	INTEGER NOT NULL DEFAULT 0
explanation	TEXT
notes	TEXT
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

Relationships: collection_id → remittance_collections.id; bus_id → buses.id

Drivers & payroll

driver_advances

Introduced in 016 — 016_create_driver_finances.sql

Column	Definition (from CREATE TABLE)
id	INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY
employment_id	INTEGER NOT NULL
advance_date	DATE NOT NULL
financial_transaction_id	INTEGER
reason	TEXT
notes	TEXT
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

Relationships: employment_id → driver_employments.id; financial_transaction_id → financial_transactions.id

driver_deductions

Introduced in 016 — 016_create_driver_finances.sql

Column	Definition (from CREATE TABLE)
id	INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY
employment_id	INTEGER NOT NULL
deduction_date	DATE NOT NULL
reason	TEXT NOT NULL
applied	BOOLEAN NOT NULL DEFAULT FALSE
notes	TEXT
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

Relationships: employment_id → driver_employments.id

driver_salaries

Introduced in 017 — 017_create_driver_salaries.sql

Column	Definition (from CREATE TABLE)
id	INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY
employment_id	INTEGER NOT NULL
salary_year	INTEGER NOT NULL
salary_month	INTEGER NOT NULL
payment_date	DATE
financial_transaction_id	INTEGER

notes	TEXT
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

Relationships: employment_id → driver_employments.id; financial_transaction_id → financial_transactions.id

salary_advance_recoveries

Introduced in 019 — 019_add_integrity_and_indexes.sql

Column	Definition (from CREATE TABLE)
id	INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY
salary_id	INTEGER NOT NULL
advance_id	INTEGER NOT NULL
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

Relationships: salary_id → driver_salaries.id; advance_id → driver_advances.id

salary_deduction_allocations

Introduced in 019 — 019_add_integrity_and_indexes.sql

Column	Definition (from CREATE TABLE)
id	INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY
salary_id	INTEGER NOT NULL
deduction_id	INTEGER NOT NULL
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

Relationships: salary_id → driver_salaries.id; deduction_id → driver_deductions.id

Maintenance & reminders

maintenance

Introduced in 015 — 015_create_maintenance.sql

Column	Definition (from CREATE TABLE)
id	INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY
bus_id	INTEGER NOT NULL
description	TEXT NOT NULL
reported_date	DATE NOT NULL
start_date	DATE
completed_date	DATE
expense_id	INTEGER
downtime_start	DATE
downtime_end	DATE
notes	TEXT
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

Relationships: bus_id → buses.id; expense_id → expenses.id

maintenance_schedules

Introduced in 015 — 015_create_maintenance.sql

Column	Definition (from CREATE TABLE)
id	INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY
bus_id	INTEGER NOT NULL
interval_days	INTEGER NOT NULL DEFAULT 35
last_completed_date	DATE
next_due_date	DATE
active	BOOLEAN NOT NULL DEFAULT TRUE
notes	TEXT
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

Relationships: bus_id → buses.id

reminders

Introduced in 018 — 018_create_reminders.sql

Column	Definition (from CREATE TABLE)
id	INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY
description	TEXT
due_date	DATE NOT NULL
bus_id	INTEGER
maintenance_schedule_id	INTEGER
business_loan_id	INTEGER
completed_date	DATE
notes	TEXT
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

Relationships: bus_id → buses.id; maintenance_schedule_id → maintenance_schedules.id; business_loan_id → business_loans.id

4. Understanding the principal workflows

Bus and driver records

A bus belongs to a route. Drivers are people, employments describe the period they work for the business, and driver assignments connect an employment to a bus over a date range. Historical records are retained rather than overwritten.

Cashing and account balance

A remittance_collection describes an occasion on which money is received; its remittances describe contributions from individual buses. A linked inbound financial_transaction records the same total in the selected financial account. Expected amounts are derived from the stored day counts/rates; variance explains differences from actual cash.

Expenses and owner-paid costs

An expense describes what was purchased or paid for and which bus(es) or general business activity it relates to. BUSINESS_ACCOUNT expenses must match an outgoing ledger movement. OWNER_PERSONAL expenses are paid with the owner's money, so they create an owner-loan obligation instead of a business-account cash movement.

Transfers between accounts

The transfer record describes an outgoing account, incoming account, amount, fee and type. Matched ledger movements represent the source outflow and destination inflow, plus any fee charged. An external transfer may have different accounting treatment; use the deferred validator as authority.

Borrowing and repayment

Business borrowing/repayments and owner funding are tracked separately. When the owner directly pays an outside business lender, the debt is effectively transferred from that lender to the owner without pretending that business cash changed hands.

Driver salary and advances

Advance payments are recorded and connected to outgoing financial entries. Salaries calculate the payable net amount after recoveries and deductions. Allocation tables assign recoveries to specific advances/deductions, preventing an advance being recovered twice or against the wrong driver.

Maintenance and reminders

Maintenance is an operational event, not automatically an expense. It can refer to a corresponding expense when money is spent, but a complimentary check can have no expense. Maintenance schedules and reminders help track future work.

5. Constraints, indexes, triggers and integrity guarantees

The database uses conventional primary keys, foreign keys, uniqueness checks, controlled status/type values, valid date/amount checks, selective unique indexes and deferred cross-table validation. Deferred checks allow one application transaction to insert interconnected records in any appropriate order, then validate their final combined state when constraints are forced immediate or at COMMIT.

Migration 022 verification functions

- sf_suggest_transfer_fee — suggest transfer fee
- sf_is_external_transfer — is external transfer
- sf_assert_transfer — Validate transfer
- sf_assert_expense — Validate expense
- sf_assert_collection — Validate collection
- sf_assert_advance — Validate advance
- sf_assert_deduction — Validate deduction
- sf_assert_salary — Validate salary
- sf_assert_owner_loan_transaction — Validate owner loan transaction
- sf_assert_owner_balance — Validate owner balance
- sf_assert_loan_repayment — Validate loan repayment
- sf_assert_business_loan — Validate business loan
- sf_assert_employment — Validate employment
- sf_assert_maintenance — Validate maintenance
- sf_assert_ledger — Validate ledger
- sf_assert_rate — Validate rate
- sf_assert_fee_rule — Validate fee rule
- sf_integrity_deferred — integrity deferred
- sf_preserve_account_type — preserve account type
- sf_touch_updated_at — touch updated at

Examples of database-enforced invariants

- Cashing collection amount must reconcile with individual bus remittances and inbound account ledger.
- Business-funded expenses must match an outgoing ledger item of the correct amount, date and account.

- Owner-funded expenses require matching OWNER_FUNDED evidence and must not fabricate a cash outflow from a business account.
- Transfers require matching entries for movement between accounts and any fee.
- Paid driver salaries must reconcile with the matching payment and recovery allocation records.
- Recovery allocations must belong to the same employment and cannot exceed an advance or deduction.
- Business loan repayments cannot exceed the outstanding loan amount.
- Maintenance cannot be attached to an expense belonging to a different bus; no-cost events need no expense.
- Configured fee bands and effective cashing rates cannot overlap improperly.
- Employment and bus assignment periods/status must be compatible.
- updated_at is automatically touched by a trigger for supported tables.

6. Migration history

001_create_routes.sql — Create routes.

002_create_buses.sql — Create buses.

003_create_drivers.sql — Create drivers.

004_create_driver_employments.sql — Create driver employments.

005_create_driver_assignments.sql — Create driver assignments.

006_create_financial_accounts.sql — Create financial accounts.

007_create_financial_transactions.sql — Create financial transactions.

008_create_transfers.sql — Create transfers.

009_seed_transfer_fee_rules.sql — Seed transfer fee rules.

010_create_expenses.sql — Create expenses.

011_create_expense_allocations.sql — Create expense allocations.

012_create_owner_loans.sql — Create owner loans.

013_create_business_loans.sql — Create business loans.

014_create_remittances.sql — Create remittances.

015_create_maintenance.sql — Create maintenance.

016_create_driver_finances.sql — Create driver finances.

017_create_driver_salaries.sql — Create driver salaries.

018_create_reminders.sql — Create reminders.

019_add_integrity_and_indexes.sql — Add integrity and indexes.

020_seed_initial_configuration.sql — Seed initial configuration.

021_add_cashing_rate_configuration.sql — Add cashing rate configuration.

022_smartfleet_schema_v1_hardening_corrected.sql — Schema V1 hardening: additional structural checks, matching-ledger validators, deferred constraint triggers, fee/rate overlap and timestamp protection.

7. Validation evidence

Before the automated run, ten manual integrity scenarios passed, covering remittance totals, transfer reconciliation, expense/ledger matching and unsupported owner funding. The corrected automated script then reported 18 tests, 18 passes and 0 failures, with a final ROLLBACK. An initial version reported 16/18 because of test-fixture dependencies; it was corrected and re-run. These results are evidence of the tested invariants, not proof of exhaustive correctness or security.

#	Scenario	Result
01	Owner personal expense + loan	PASS
02	Unbacked owner repayment rejected	PASS
03	Matched owner deposit	PASS
04	Overpaid business loan rejected	PASS
05	Owner pays business lender	PASS
06	Matching driver advance	PASS
07	Mismatched driver advance rejected	PASS
08	Missing salary recovery allocation rejected	PASS
09	Cross-driver recovery rejected	PASS
10	Matching paid payroll	PASS
11	Wrong-bus maintenance expense rejected	PASS
12	No-cost completed maintenance	PASS
13	Automatic updated_at	PASS
14	Overlapping fee rule rejected	PASS
15	Overlapping cashing rate rejected	PASS
16	Fee suggestion calculations	PASS
17	Terminated assignment rejected	PASS
18	Advance over-recovery rejected	PASS

8. Implementation guidance and operating practices

Keep operations atomic

Create a single application transaction for business operations involving several tables. Insert the necessary records; validate the final state; commit only when every component succeeds. Roll back on exceptions.

Separate the API from the database

Streamlit may initially call a Python service/repository layer, but avoid writing business logic only in UI handlers. A future FastAPI endpoint can reuse the same domain operations and database rules.

Calculate rather than duplicate

Use generated columns and server-side validations as the authoritative result for net salary, cashing variance and financially significant relationships. Do not rely only on UI arithmetic.

Accountability

Use least-privilege database roles, prepared statements/parameterised queries, environment-managed secrets, secure passwords and tested backups before live deployment. Authentication and access policies remain a separate application/security phase.

Versioning and changes

Treat migrations as append-only history after application. Never casually re-run 022 on a database where it has already been committed; introduce a new numbered migration for future schema changes. Keep corrected source files with one canonical filename.

Testing

Rerun the test suite in a disposable development/test database after every migration, and add tests for every discovered defect. The suite intentionally ROLLBACKs; do not use it as a live-data import.

Performance

Monitor query plans before adding indexes indiscriminately. Pay particular attention to date-filtered financial reports, current bus assignments, loan balances and salary reporting.

9. Scope boundaries and open decisions

- The MVP does not require recording individual journeys or tracking fuel consumption.
- No-cost maintenance is permitted; maintenance records are not forcibly treated as payments.
- The schema records operational/financial data; it is not by itself a complete authentication, authorization or audit-log solution.
- Business policy should still settle reporting definitions (gross vs net), company-wide versus bus expenses, and operational permissions.
- Actual application end-to-end tests, concurrent-session tests, backup/restore drills and permissions testing remain future work.

Appendix A — Foreign-key relationship register

Dependent table.column	References
buses.route_id	routes.id
driver_advances.employment_id	driver_employments.id
driver_advances.financial_transaction_id	financial_transactions.id
driver_assignments.bus_id	buses.id
driver_assignments.employment_id	driver_employments.id
driver_deductions.employment_id	driver_employments.id
driver_employments.driver_id	drivers.id
driver_salaries.employment_id	driver_employments.id
driver_salaries.financial_transaction_id	financial_transactions.id
expense_allocations.bus_id	buses.id
expense_allocations.expense_id	expenses.id
expenses.bus_id	buses.id
expenses.category_id	expense_categories.id
expenses.financial_account_id	financial_accounts.id
expenses.financial_transaction_id	financial_transactions.id
financial_transactions.account_id	financial_accounts.id
financial_transactions.transfer_id	transfers.id
loan_repayments.financial_transaction_id	financial_transactions.id
loan_repayments.loan_id	business_loans.id
loan_repayments.owner_loan_transaction_id	owner_loan_transactions.id
maintenance.bus_id	buses.id
maintenance.expense_id	expenses.id
maintenance_schedules.bus_id	buses.id
owner_loan_transactions.expense_id	expenses.id

owner_loan_transactions.financial_transaction_id	financial_transactions.id
owner_loan_transactions.owner_loan_id	owner_loans.id
reminders.bus_id	buses.id
reminders.business_loan_id	business_loans.id
reminders.maintenance_schedule_id	maintenance_schedules.id
remittance_collections.financial_account_id	financial_accounts.id
remittance_collections.financial_transaction_id	financial_transactions.id
remittances.bus_id	buses.id
remittances.collection_id	remittance_collections.id
salary_advance_recoveries.advance_id	driver_advances.id
salary_advance_recoveries.salary_id	driver_salaries.id
salary_deduction_allocations.deduction_id	driver_deductions.id
salary_deduction_allocations.salary_id	driver_salaries.id
transfers.destination_account_id	financial_accounts.id
transfers.fee_rule_id	transfer_fee_rules.id
transfers.source_account_id	financial_accounts.id

Appendix B — Provenance

Compiled from the uploaded migrations.zip containing migrations 001–022 and smartfleet_integrity_tests_corrected.sql. This is source-derived documentation, not a fresh introspection of a running PostgreSQL instance. The applied database and source migrations should remain in sync.